

PROJECT REPORT

RETAIL SALES & PROFIT PERFORMANCE ANALYTICS



SQL Server Data Cleaning & Analysis + Tableau Visualization

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1. Project Overview

This project analyzes retail sales and profit data using SQL Server for data inspection, cleaning,

updating, structural/data preparation, and analytical querying, followed by Tableau visualization. The final dashboard presents sales, profit, customers, orders, customer performance, monthly sales trends, category analysis, and discount-related profit analysis.

2. Project Objectives

- Inspect the structure and contents of the retail table before analysis.
- Identify null values and potential duplicate Order_ID records.
- Update missing or invalid values where required.
- Use SQL aggregate functions for business-level summaries.
- Use SQL window functions for analytical calculations.
- Use filtering and grouping techniques such as GROUP BY and HAVING.
- Prepare analytical outputs for Tableau.
- Build an interactive dashboard for sales and profit performance analysis.

3. Tools & Technologies

Tool	Role	Evidence
SQL Server	Data inspection, cleaning, updating, and analysis	SQL query screenshots
SQL	SELECT, UPDATE, GROUP BY, HAVING, aggregate, and window-based analysis	SQL workflow
Tableau	Visualization and interactive dashboarding	Final dashboard

4. Dataset Fields Visible in SQL Server

The SQL Server result grid shows these fields in the Superstore_sql table:

- Order_ID
- Order_Date
- Customer_Name
- Segment
- Region
- Category
- Product_Name
- Sales
- Quantity
- Discount
- Profit

The result grid contains order records with customer, segment, region, category, product, sales, quantity, discount and profit values.

5. SQL Server Data Preparation & Cleaning

The shown workflow starts with retrieving Superstore_sql, inspecting its schema, checking null values, updating missing values, and checking duplicate Order_ID values.

5.1 Data Retrieval

- SELECT * FROM Superstore_sql was used to retrieve and inspect the table data.
- The result grid was used to review the available fields and records.

The screenshot shows the Tableau SQL Editor interface. The query editor at the top contains the following SQL code:

```
USE SQLTABLEAU
SELECT * FROM Superstore_sql
```

The status bar indicates "Query executed successfully." Below the query editor, the "Results" pane displays a grid of 21 order records. The columns are: Order_ID, Order_Date, Customer_Name, Segment, Region, Category, Product_Name, Sales, Quantity, Discount, and Profit.

Order_ID	Order_Date	Customer_Name	Segment	Region	Category	Product_Name	Sales	Quantity	Discount	Profit
ORD-1001	2023-11-24	Vishal Singh	Consumer	Central	Furniture	Product-15	504.65	2	0.00	457.64
ORD-1002	2023-01-17	Aarav Sharma	Consumer	West	Furniture	Product-33	1945.35	3	0.10	636.83
ORD-1003	2023-12-26	Shruti Kulkarni	Corporate	West	Office Supplies	Product-38	996.75	1	0.10	607.58
ORD-1004	2023-06-24	Dipa Mahapatra	Consumer	West	Office Supplies	Product-7	368.96	2	0.20	801.76
ORD-1005	2023-11-06	Dipa Mahapatra	Consumer	South	Technology	Product-8	2922.04	7	0.00	417.65
ORD-1006	2023-11-18	Manish Tiwari	Corporate	West	Technology	Product-5	232.89	4	0.20	986.79
ORD-1007	2023-04-30	Vishal Singh	Corporate	Central	Office Supplies	Product-24	571.70	6	0.10	571.23
ORD-1008	2023-12-26	Deepak Mahra	Home Office	East	Technology	Product-11	1649.01	4	0.10	306.34
ORD-1009	2023-05-19	Sumit Yadav	Home Office	West	Technology	Product-21	2544.27	1	0.10	768.34
ORD-1010	2023-06-11	Rahul Khanna	Corporate	East	Furniture	Product-37	2641.47	6	0.10	552.07
ORD-1011	2023-07-22	Sumit Yadav	Corporate	West	Office Supplies	Product-9	815.22	3	0.20	671.12
ORD-1012	2023-08-08	Karan Mahapatra	Corporate	Central	Furniture	Product-9	1577.63	2	0.00	819.43
ORD-1013	2023-03-20	Sumit Yadav	Consumer	South	Technology	Product-5	1215.83	10	0.30	387.85
ORD-1014	2023-10-11	Aarav Sharma	Home Office	East	Technology	Product-35	2277.55	6	0.00	81.55
ORD-1015	2023-03-22	Anil Agarwal	Consumer	Central	Technology	Product-12	1572.28	2	0.20	794.17
ORD-1016	2023-09-17	Manish Tiwari	Consumer	West	Office Supplies	Product-11	1664.20	3	0.00	478.63
ORD-1017	2023-09-08	Aarav Sharma	Consumer	Central	Office Supplies	Product-15	267.58	10	0.00	-188...
ORD-1018	2023-08-06	Aditya Verma	Home Office	West	Furniture	Product-31	2845.96	3	0.20	305.56
ORD-1019	2023-11-07	Siddhant Jain	Consumer	West	Technology	Product-30	1257.08	6	0.30	809.50
ORD-1020	2023-08-20	Vishal Singh	Consumer	West	Furniture	Product-22	161.80	9	0.10	464.97
ORD-1021	2023-03-04	Aditya Verma	Home Office	East	Furniture	Product-5	2725.72	6	0.00	362.41

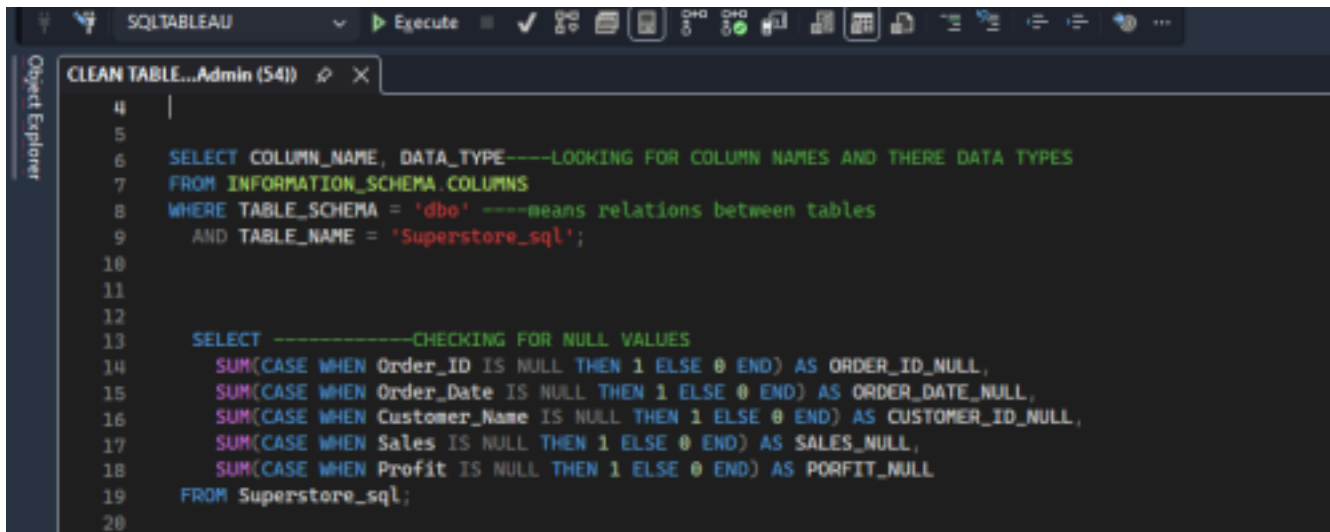
Figure 1. SQL Server table data retrieved from Superstore_sql

5.2 Column & Data-Type Inspection

- INFORMATION_SCHEMA.COLUMNS was queried to inspect column names and data types.
- The query was filtered to the dbo schema and Superstore_sql table.

5.3 Null-Value Checking

The screenshot shows SUM(CASE WHEN ... IS NULL THEN 1 ELSE 0 END) logic to count missing values for Order_ID, Order_Date, Customer_Name, Sales, and Profit.



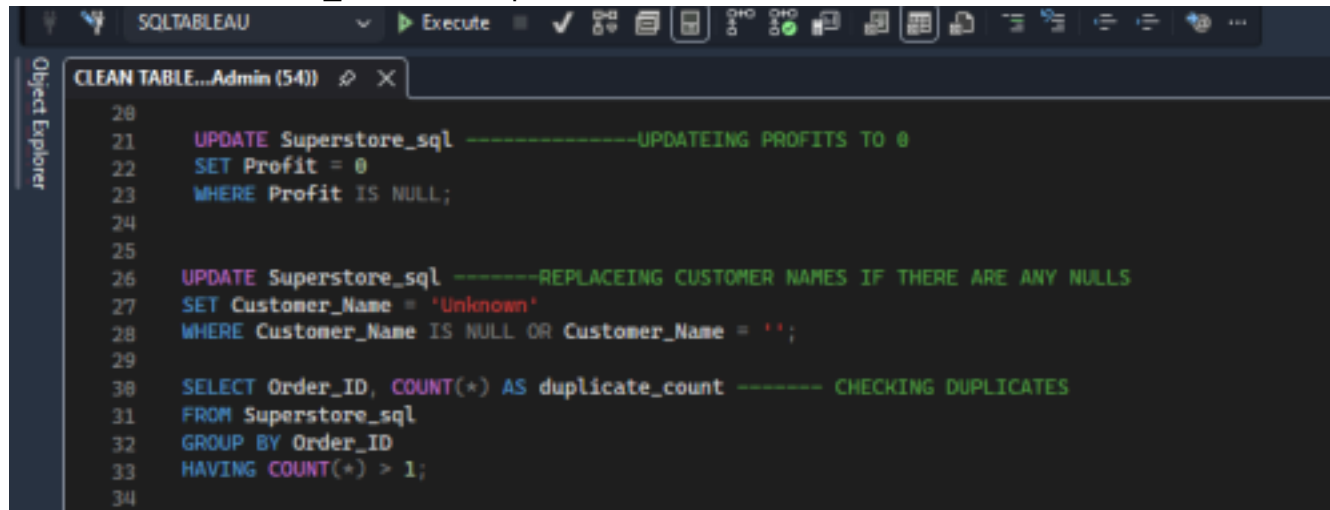
```

4
5
6 SELECT COLUMN_NAME, DATA_TYPE-----LOOKING FOR COLUMN NAMES AND THERE DATA TYPES
7 FROM INFORMATION_SCHEMA.COLUMNS
8 WHERE TABLE_SCHEMA = 'dbo' -----means relations between tables
9      AND TABLE_NAME = 'Superstore_sql';
10
11
12
13 SELECT -----CHECKING FOR NULL VALUES
14      SUM(CASE WHEN Order_ID IS NULL THEN 1 ELSE 0 END) AS ORDER_ID_NULL,
15      SUM(CASE WHEN Order_Date IS NULL THEN 1 ELSE 0 END) AS ORDER_DATE_NULL,
16      SUM(CASE WHEN Customer_Name IS NULL THEN 1 ELSE 0 END) AS CUSTOMER_ID_NULL,
17      SUM(CASE WHEN Sales IS NULL THEN 1 ELSE 0 END) AS SALES_NULL,
18      SUM(CASE WHEN Profit IS NULL THEN 1 ELSE 0 END) AS PORFIT_NULL
19 FROM Superstore_sql;
20
  
```

Figure 2. Null-value checking using CASE and aggregate logic

5.4 Updating Missing Values

- Profit was updated to 0 where Profit IS NULL.
- Customer_Name was updated to 'Unknown' where it was NULL or blank.



```

20
21 UPDATE Superstore_sql -----UPDATEING PROFITS TO 0
22 SET Profit = 0
23 WHERE Profit IS NULL;
24
25
26 UPDATE Superstore_sql -----REPLACEING CUSTOMER NAMES IF THERE ARE ANY NULLS
27 SET Customer_Name = 'Unknown'
28 WHERE Customer_Name IS NULL OR Customer_Name = '';
29
30 SELECT Order_ID, COUNT(*) AS duplicate_count ----- CHECKING DUPLICATES
31 FROM Superstore_sql
32 GROUP BY Order_ID
33 HAVING COUNT(*) > 1;
34
  
```

Figure 3. SQL Server updates for missing Profit and Customer_Name

5.5 Duplicate Checking

Duplicate Order_ID values were checked with GROUP BY Order_ID, COUNT(*) and HAVING

COUNT(*) > 1 to identify IDs appearing more than once.

6. SQL Analysis Techniques Used

Technique	Purpose Evidence / Description
SELECT	Retrieve and inspect records. SELECT * FROM Superstore_sql
CASE	Conditional data checks NULL counting.
UPDATE	Correct missing values Profit and Customer_Name
GROUP BY	Group records Duplicate Order_ID check
HAVING	Filter grouped results. COUNT(*) > 1
Aggregate functions	Summarise data. Used for analytical summaries
Window functions	Analytical calculations while retaining row-level detail Used in the stated SQL workflow
ALTER / structural SQL	Modify table structure when required. Used in the stated SQL workflow

The screenshots do not show the exact window function or ALTER statements, so no specific function or ALTER operation is claimed here.

7. Tableau Dashboard

The Tableau dashboard combines KPI cards, customer-level analysis, a monthly sales trend, category comparison, and discount-related profit analysis. Filters are visible for the category, segment, region and quarter of the order date.

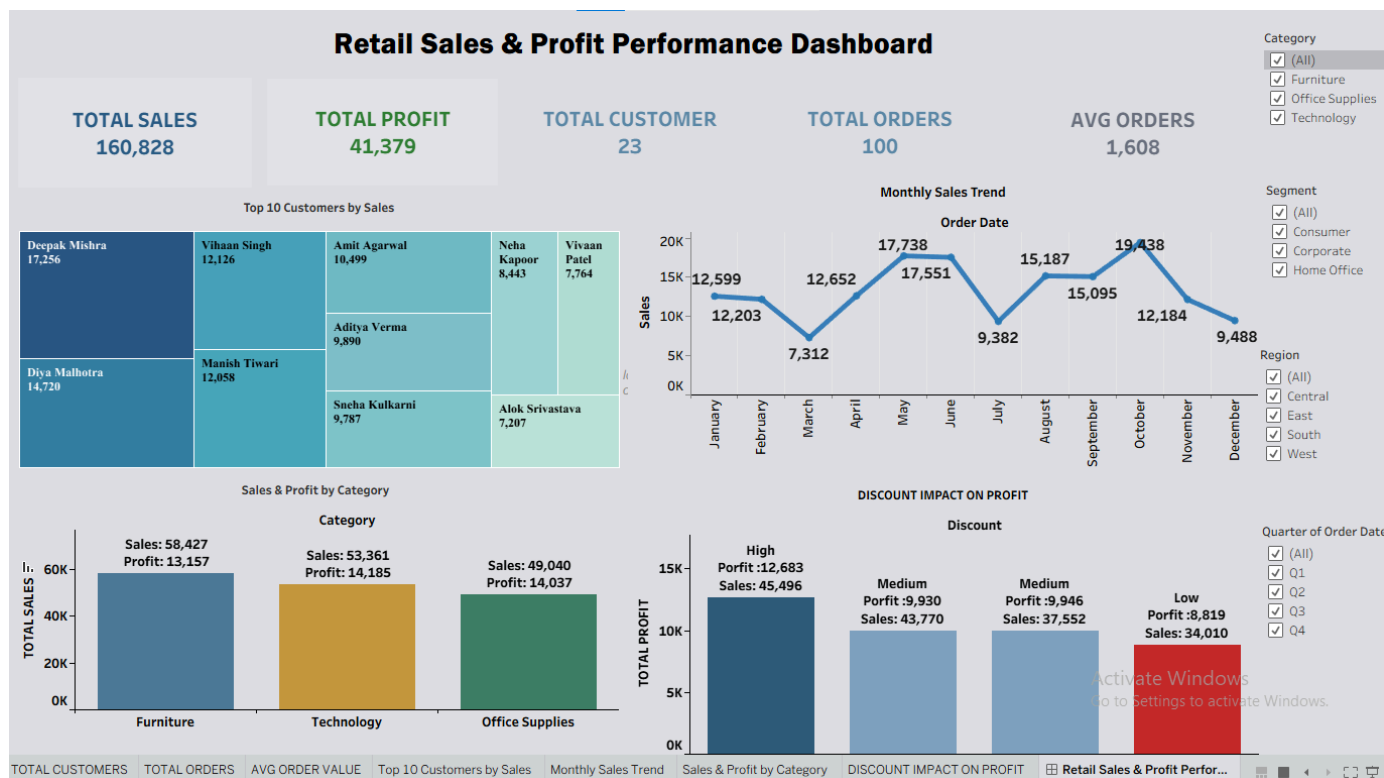


Figure 4. Final Tableau Retail Sales & Profit Performance Dashboard

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8. Dashboard Components

Dashboard Element	What It Shows
KPI Cards	Total Sales : 160,828; Total Profit: 41,379; Total Customers: 23; Total Orders: 100; Avg Orders: 1,608.
Top 10 Customers by Sales	Treemap of customer sales; Deepak Mishra is the largest displayed customer at 17,256.
Monthly Sales Trend	Line chart showing month-to-month sales values.
Sales & Profit by Category	Comparison across Furniture, Technology, and Office Supplies.
Discount Impact on Profit	Bar chart comparing profit across the displayed discount groups.

Filters Category, Segment, Region, and Quarter of Order Date.

9. Observations Supported by the Dashboard

- The dashboard reports total sales of 160,828 and total profit of 41,379.
- The dashboard displays 23 customers and 100 orders.
- The displayed average-orders KPI is 1,608.
- Deepak Mishra is the largest displayed customer in the Top 10 Customers by Sales treemap, with 17,256 in sales.
- The monthly sales line chart shows visible month-to-month variation, including a displayed high of 19,438 and a low of 7,312 among the labeled points.
- The dashboard compares furniture, technology, and office supplies.
- The dashboard includes a discount-versus-profit comparison.
- Filters are provided for Category, Segment, Region, and Quarter of Order Date.

10. End-to-End Project Workflow

1. Retrieve Superstore_sql data in SQL Server.
2. Inspect columns and data types.
3. Check missing values.
4. Update missing Profit and Customer_Name values.
5. Check duplicate Order_ID values.
6. Apply SQL analytical techniques, including aggregation and window-based analysis.
7. Prepare analytical results for visualization.
8. Build Tableau KPI, customer, trend, category, and discount visuals.
9. Add dashboard filters for interactive analysis.

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11. Business Questions Addressed

- What are the overall sales, profit, and customer and order figures?
- Which customers contribute the most sales?
- How do sales change over the displayed months?
- How are the displayed categories performing?
- How does profit vary across discount groups?
- How can the dashboard be filtered by category, segment, region, and quarter?

12. Conclusion

This project demonstrates an end-to-end retail analytics workflow using SQL Server and Tableau. SQL Server was used to inspect, clean, update, and analyze the data, while Tableau was used to transform the resulting information into an interactive sales and profit dashboard.